Real Estate Price Exploration Tool: An Exploratory Data Analysis Dashboard for Property Price Analytics

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***Abstract*—**Property prices are shaped by size, location, age, configuration, and amenities, yet buyers, sellers, students, and analysts often encounter listing data as long tables that make comparison and pattern recognition slow. This paper presents the Real Estate Price Exploration Tool, an interactive dashboard that applies exploratory data analysis (EDA) to property-listing records. The tool combines data-quality auditing, descriptive statistics, derived measures such as price per square foot and log price, Pearson and Spearman correlation, simple and log-linear regression summaries, locality-level comparison, and filterable visualizations including histograms, scatter plots, box plots, and a correlation heatmap. The methodology is grounded in hedonic pricing theory and positions the tool as a transparent, descriptive complement to predictive housing-price models rather than a replacement for them. The paper specifies the dataset schema, cleaning rules, statistical measures, dashboard modules, traceability principles, and a verification plan, and it states the interpretive limits of observational listing data: asking prices are not transaction prices, associations are not causal effects, and results describe the analyzed file only. Numerical results are reported through templates to be populated from the finalized dataset; no figures are asserted before that step.

Index Terms*—real estate, property price analysis, exploratory data analysis, hedonic pricing, price per square foot, correlation analysis, data visualization, interactive dashboard*

I. INTRODUCTION

Housing is the largest financial decision many households make, and listing portals publish thousands of records that differ in locality, built-up area, number of bedrooms, age, and amenities. A prospective buyer comparing two apartments, a student studying market structure, or an analyst scanning a city usually faces the same obstacle: the data are tabular, the relationships between variables are not visible, and a single headline figure such as the average price hides wide spread across localities and property types. Exploratory data analysis (EDA), introduced by Tukey [7], addresses this by summarizing data, exposing irregularities, and generating questions before any model is fitted.

Economic research on housing prices is dominated by the hedonic framework, in which the price of a property is understood as a bundle of implicit prices for its characteristics such as size, location, and condition [1], [3]. Machine-learning studies have since compared hedonic regressions with neural networks and ensemble methods for price prediction [5], [6]. These approaches target predictive accuracy. A complementary and often neglected need is a transparent, interactive way to *look* at the data: to see how price is distributed, how it moves with area, and how localities differ, with every displayed number traceable to a field and a formula.

This paper describes the Real Estate Price Exploration Tool, a dashboard whose analytical emphasis is descriptive rather than predictive. It integrates a data-quality audit, summary statistics, derived price measures, correlation and regression summaries, locality comparisons, and filterable charts. Its purpose is to make price patterns easier to inspect, not to assert what a property is worth or to recommend a purchase.

The contributions are: (1) a reproducible EDA workflow for property-listing data; (2) a set of derived measures and statistical procedures chosen to suit the skewed distribution of prices; (3) a dashboard design that exposes these analyses through linked views with explicit traceability; (4) a focused literature table of eight related works across hedonic pricing, price prediction, and dashboard design; and (5) a verification plan and a statement of interpretive limits. The paper does not claim that correlation, regression, or dashboards are novel in themselves, nor that the tool predicts prices.

II. RELATED WORK

The literature relevant to this project falls into four themes: (i) hedonic price theory and the composition of hedonic models, (ii) data-driven housing-price prediction, (iii) exploratory analysis and dashboard design, and (iv) data quality and responsible use. The themes are connected but answer different questions: hedonic studies estimate the contribution of attributes, prediction studies compare algorithms, and dashboard studies examine how information is presented and interpreted.

Hedonic pricing. Rosen formalized the idea that differentiated goods are priced as bundles of attributes [1]. Harrison and Rubinfeld applied this to housing, relating property values to structural and neighbourhood variables [2]. Sirmans, Macpherson, and Zietz reviewed the variables used in many hedonic models, showing that lot size, floor area, age, bedrooms, and location recur, while the sign and magnitude of an attribute depend on market and context [3]. Malpezzi gives an applied review of the method and its pitfalls, including functional form and omitted variables [10].

These works justify the variables the dashboard emphasizes but also warn that an attribute's observed association with price is not its causal value.

Price prediction. Limsombunchai compared a hedonic model with an artificial neural network for house prices [6], and Park and Bae applied several machine-learning algorithms to housing data from Fairfax County, Virginia [5]. De Cock introduced the Ames, Iowa dataset as a richer alternative to the older Boston data for teaching regression [4]. These studies evaluate predictive error on held-out data. The present tool does not train a predictor; its claims concern distributions and pairwise associations, and it should not be compared with the accuracy figures reported in this literature.

Exploration and dashboards. Tukey's EDA tradition stresses looking at data before modelling [7], and Anscombe showed that identical summary statistics can conceal very different data structures, which is why scatter plots accompany every correlation coefficient [8]. Sarikaya et al. analyzed how the term "dashboard" is used and identified design purposes ranging from decision support to communication [9], and Verbert et al. surveyed dashboard applications and open research directions [11]. Together these works support combining summary cards, distributions, relationships, and group comparisons in one interface, while indicating that building a dashboard is not evidence that users decide better.

Data quality and responsible use. Listing data are asking prices entered by sellers or agents, often with rounded values, missing fields, duplicated advertisements, and inconsistent locality names. Analyses built on such data should audit quality first, document provenance, and avoid presenting asking-price statistics as market valuations.

TABLE I LITERATURE SURVEY: EIGHT CLOSELY RELATED WORKS

Ref.	Theme	Focus / method	Relevance to this tool
[1] Rosen, 1974	Hedonic theory	Prices of differentiated goods as attribute bundles	Justifies analyzing price against property attributes
[2] Harrison & Rubinfeld, 1978	Hedonic housing	Housing values vs. structural and neighbourhood variables	Early model for attribute–price analysis
[3] Sirmans et al., 2005	Hedonic review	Variables used across hedonic models	Guides choice of area, age, bedrooms, location
[4] De Cock, 2011	Dataset	Ames housing data for regression teaching	Shows value of a documented housing dataset
[5] Park & Bae, 2015	ML prediction	Algorithms for housing price prediction	Predictive contrast; tool stays descriptive
[6] Limsombunchai, 2004	Hedonic vs. ANN	Compares regression with neural network	Motivates a transparent baseline
[9] Sarikaya et al., 2019	Dashboards	Typology of dashboard purposes	Informs view layout and intended use
[11] Verbert et al., 2014	Dashboards	Overview of dashboard applications	Design tradition applied to listing data

A. Synthesis and Research Gap

Hedonic and machine-learning studies estimate or predict prices; dashboard studies discuss presentation. The gap addressed here is practical integration: a compact tool that places data-quality checks, derived price measures, correlations, and locality comparisons in one interface, and that states what each display can and cannot support. The gap should not be read as an absence of property portals or price-prediction research. The contribution is the applied workflow and its explicit reporting of provenance, definitions, and limits.

III. PROBLEM STATEMENT AND RESEARCH QUESTIONS

Raw listing tables make it difficult to inspect how price varies with area, configuration, age, and locality at the same time. The Real Estate Price Exploration Tool aims to: (1) audit and summarize the supplied listing data; (2) examine the distributions of price, area, and price per square foot; (3) estimate linear and rank associations between price and property attributes; (4) compare localities and property types; and (5) present these summaries in an interactive interface that makes uncertainty and non-causal interpretation explicit.

TABLE II RESEARCH QUESTIONS AND BOUNDARIES

ID	Research question	Boundary of the answer
RQ1	What are the distributions and quality characteristics of the listing dataset?	Describes this file only; representativeness unknown
RQ2	How are area, bedrooms, and age associated with price and price per square foot?	Association, not causation; omitted variables
RQ3	How do price levels differ across localities and property types?	Exploratory; listing mix differs by locality
RQ4	How clearly does each dashboard view map to a reproducible calculation?	Traceability can be inspected; usability is not measured
RQ5	What must be resolved before outputs inform consequential decisions?	Produces an evaluation agenda, not a valuation

A. Conceptual Framework and Non-Claims

The project follows an evidence chain: raw records are treated as data requiring inspection; quality checks determine what can be computed; statistics summarize the file; visualizations support human inspection; and navigation supports workflow. None of these steps independently establishes causality or the fair value of a property. The tool combines descriptive analytics and bivariate association analysis. Predictive analytics would require a defined target, temporally valid predictors, held-out evaluation, and validation in another market; prescriptive analytics would require a justified mapping from evidence to action. Neither is claimed. Accordingly the paper uses terms such as "observed association" and "descriptive pattern" rather than "cause," "valuation," or "price prediction."

IV. SYSTEM OVERVIEW AND METHODOLOGY

A. Workflow and Architecture

Figure 1 shows the workflow. A listing file is ingested, audited, cleaned, enriched with derived measures, analyzed statistically, and rendered in the interface. Logically the

system has four layers: data input (file or approved feed, with a contract of field names, types, ranges, and provenance); analytics (descriptive summaries, associations, group aggregates); presentation (charts, cards, tables, filters); and governance (versioning, data minimization, permissions, documentation, review). This layering is a design model for the prototype, not a claim that every control is already implemented.

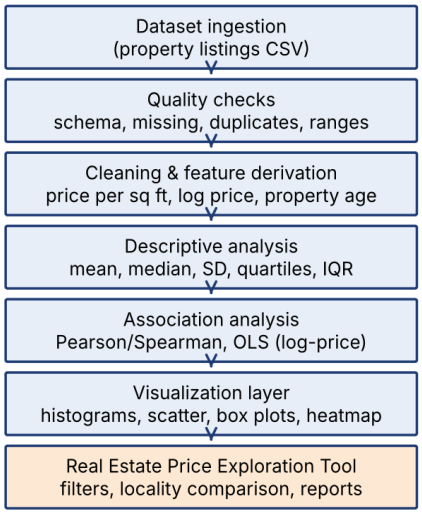


Fig. 1. Analysis and presentation workflow of the Real Estate Price Exploration Tool. The diagram shows the reported workflow, not a production data pipeline.

B. Dataset and Schema

The tool expects a property-listing table with one row per advertised property. Table III lists the expected fields and their analytical treatment. The dataset name, source, collection date, and licence must be recorded in the final manuscript; this version specifies the schema so that the analysis can be reproduced once the file is fixed. Field meanings follow the column names and are not independently verified against a source data dictionary.

TABLE III EXPECTED FIELDS AND ANALYTICAL TREATMENT

Field	Type	Treatment
listing_id	Identifier	Duplicate check; not an analytical variable
price	Numeric (INR)	Primary outcome; right-skewed, so log-transformed
area_sqft	Numeric	Primary attribute; basis for price per sq ft
bedrooms, bathrooms	Integer	Configuration attributes; summarized and correlated
age_years	Numeric	Property age; correlated with price
locality / city	Category	Group counts, medians, box plots
property_type	Category	Apartment, house, villa, plot; group comparison
furnishing, amenities	Category/flag	Descriptive counts; used as filters
listing_date	Date	Schema inspection; optional time filter

C. Data-Quality Audit and Cleaning Rules

Before any statistic is computed the tool reports: row and column counts; missing values per field; exact duplicate rows and duplicate identifiers; near-duplicate advertisements (same locality, area, and price); numeric ranges; and category spellings. Cleaning rules are explicit and logged rather than silent. Records with non-positive price or area are flagged. Locality names are trimmed and case-normalized, and the mapping applied is stored. Missing values are not imputed in the primary analysis; affected rows are excluded pairwise from the relevant statistic and the exclusion count is displayed. Extreme values are flagged using the interquartile-range rule (below Q1 − 1.5 IQR or above Q3 + 1.5 IQR) but are not deleted automatically, because high-value properties may be genuine. A filter toggle lets the user view results with and without flagged records.

D. Derived Measures

Raw price is strongly right-skewed in most markets, so the tool derives measures that support fair comparison across property sizes:

$$price\ per\ sq\ ft = price / area_sqft \quad (1)$$

$$log\ price = ln(price) \quad (2)$$

Price per square foot normalizes for size and is the main basis of locality comparison. The log transformation reduces skew and lets a regression slope be read approximately as a proportional change in price per unit change in the attribute. Property age is computed from a construction year where available.

E. Statistical Measures

For each numeric variable the tool reports mean, standard deviation, median, quartiles, minimum, and maximum; because prices are skewed, medians and quartiles are displayed beside means. For paired observations (x_i, y_i), Pearson's coefficient is

$$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2]} \quad (3)$$

Pearson's r captures linear association and is sensitive to outliers; Spearman's rank coefficient is therefore reported beside it as a robust check for monotonic relationships. Approximate 95% confidence intervals use Fisher's transformation [12], $z = \operatorname{atanh}(r)$, with standard error $1/\sqrt{(n-3)}$, transformed back to the correlation scale. Two-sided p-values are descriptive; no multiplicity correction is applied because the analysis is exploratory.

For each primary pair, ordinary least squares fits a line of the form $\hat{y} = \beta_0 + \beta_1 x$, with log price as the response in the log-linear specification:

$$\widehat{ln(price)} = \beta_0 + \beta_1 \cdot ln(area_sqft) \quad (4)$$

The slope β_1 then approximates the elasticity of price with respect to area within the file. The coefficient of determination R^2 is an in-sample measure of variance represented by the fitted line, not out-of-sample accuracy. Residual assumptions, measurement error, confounding, and

independence of listings are not established, so slopes describe fitted lines and must not be read as the value of an additional square foot.

F. Banded Summaries

To support communication, area and age are grouped into descriptive bands (for example, area at most 800, 801–1,200, 1,201–1,800, and above 1,800 sq ft). The bands are readability choices, not market standards or validated categories. Binning loses information and creates artificial boundaries, so banded medians are always shown beside continuous scatter plots. Band membership is not used to train a model or to label any property as over- or under-priced.

G. Dashboard Modules and Requirements

Figure 2 shows the intended layout: a filter panel, summary cards, and four linked analytical panels. Table IV records the functional requirements and the evidence needed to verify each. These form a test plan; a visible chart shows that a component exists, but numerical correctness requires comparison with a reproducible reference calculation and usability requires evaluation with representative users.

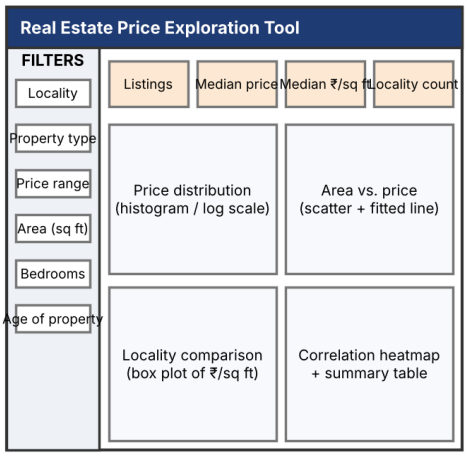


Fig. 2. Wireframe of the proposed dashboard layout (design sketch, not a screenshot of results).

TABLE IV FUNCTIONAL REQUIREMENTS AND VERIFICATION EVIDENCE

ID	Requirement	Acceptance evidence
FR-01	Summary cards: listings, median price, median price per sq ft	Recompute from a versioned file within a stated rounding tolerance
FR-02	Distributions of price, area, and price per sq ft	Compare plotted bins and counts with independently generated figures
FR-03	Scatter plots with fitted lines and correlation heatmap	Recompute r, slope, intercept, and R^2 from the same data version
FR-04	Locality and property-type comparison	Verify labels, group sizes, medians, and spread
FR-05	Filters (locality, type, price, area, bedrooms, age)	Filtered view matches a manual subset; active filters visible
FR-06	Data-quality panel with outlier toggle	Counts of missing, duplicate, and flagged rows match audit script
FR-07	Exportable summary report	Exported values equal on-screen values; filters and data version recorded

H. Analytical Traceability

Traceability means that every displayed quantity can be associated with a source field, a formula, a filter or grouping rule, and a dataset version. It matters because several views draw on the same data: a locality median, an overall median, and a card total must use consistent row inclusion rules. When a filter excludes observations, the filter is shown on screen. The minimum record stored with an exported report is a data-version identifier, computation timestamp, metric name, source field, and calculation rule. A fuller release process adds a small fixture dataset with known values, automated tests of each metric function, and a parity report comparing the dashboard with an offline script.

I. Reproducibility Protocol

The analysis follows a fixed sequence so that another researcher can repeat it: (i) load the file preserving header names; (ii) check counts, missing values, duplicates, and identifier uniqueness; (iii) apply logged cleaning rules; (iv) derive price per square foot and log price; (v) summarize numeric columns; (vi) compute Pearson and Spearman coefficients with confidence intervals for pre-specified pairs; (vii) fit separate simple regressions; (viii) aggregate by locality, property type, and band; and (ix) generate all tables and figures from the same in-memory data frame, for example with pandas [13]. A release-quality record should include the file name and checksum, library versions, the analysis command, the dashboard release identifier, and generated logs.

J. Diagnostics and Robustness Checks

Because simple regression and correlation rest on assumptions that listing data often violate, the tool provides optional diagnostics rather than presenting fitted lines as settled. Residuals from the log-linear fit are plotted against fitted values to reveal curvature or changing spread (heteroscedasticity); a pattern in this plot indicates that a straight line is an incomplete description. Pearson and Spearman coefficients are compared, and a marked difference between them signals that outliers or a non-linear relationship are driving the linear coefficient. Results are recomputed with and without IQR-flagged records, and a material change is reported. Where sample sizes allow, associations are also computed within property types, since a pooled correlation can arise from differences between groups (apartments versus villas, for example) rather than from a relationship inside each group.

These checks are descriptive aids. They do not turn an observational file into a causal design, and they are not a substitute for a pre-specified modelling protocol. If a predictive model is added later, residual analysis, cross-validation, and error reporting by locality would form part of a separate evaluation.

K. Descriptive Versus Predictive Pipelines

It is useful to state how this tool differs from a price-prediction pipeline. A predictive pipeline defines a target, splits data into training and test sets, tunes a model, and reports error on withheld records; its output is an estimate for a new property. The exploration tool performs none of these steps. Its inputs are the full file, its outputs are summaries and plots, and its evaluation concerns whether the displayed numbers are correct and interpretable. Mixing the two would risk quoting in-sample R^2 as if it were predictive accuracy, or presenting a fitted line as a price estimate. The interface therefore avoids any "estimated value" field.

V. DASHBOARD DESIGN AND USE SCENARIOS

A. Mapping Views to Computations

Each dashboard view is tied to a class of computation so that the interface can be checked against an offline reference. Table V lists the views, the analytical content each displays, and the interpretation that is and is not supported. The table separates what can be recomputed directly from the data from what would need user evaluation.

TABLE V DASHBOARD VIEW-TO-ANALYSIS MAPPING

View	Displayed content	Interpretation limit
Overview	Listing count, median price, median price per sq ft, locality count	Summaries of the file; not market indices
Data quality	Missing, duplicate, and flagged-outlier counts; range checks	Quality of this file only
Distributions	Histograms of price, area, price per sq ft (linear and log scale)	Shape and spread; not a valuation
Relationships	Scatter plots with fitted lines; correlation heatmap	Association, not causation
Localities	Box plots and medians of price per sq ft with group sizes	Exploratory; mix of properties differs
Reports	Summary tables and export with data version and filters	Consistent with on-screen values

B. Visualization Design Principles

Chart choices follow the question being asked. A histogram shows the shape of the price distribution and makes skew visible; a log-scaled axis is offered as a toggle because prices spanning an order of magnitude compress most records into one corner of a linear plot. A scatter plot with a fitted line is used for two continuous variables, and the fitted line is drawn behind semi-transparent points so that overplotting in dense regions remains visible. Box plots compare price per square foot across localities because they display median, quartiles, and flagged extremes at once, and each box is labelled with its sample size. A diverging colour scale centred on zero is used for the correlation heatmap so that sign and strength can be read together, and every cell shows its numeric value, because colour alone is not an accurate encoding of magnitude and is inaccessible to some users.

Interaction follows the same cautious logic. Filters are linked, so a change to locality or price range updates every panel, and the active filters and the resulting record count are always visible. Hovering reveals the underlying value, count, and definition. Outlier handling is user-controlled rather than hidden. Defaults are chosen to avoid misleading displays: axes start from the data range with labels, tables state units, and any chart based on fewer than a minimum number of records carries a visible warning.

C. Intended Users and Scenarios

The tool is intended for three groups. A prospective buyer or seller can compare the distribution of price per square foot for a chosen property type in several localities, and see how wide the spread is before reading any single advertisement. A student or analyst can examine how strongly area, bedrooms, and age are associated with price and observe how the choice of a log scale changes the picture. An instructor or reviewer can use the data-quality and traceability features to check how each number was produced. These scenarios are design intents inferred from the project aim; no formal observation of users has been performed.

Three scenarios illustrate the reading of outputs. In the first, a locality shows a high median price per square foot but only a handful of listings; the tool labels the sample size and the user is guided not to treat the median as stable. In the second, area and price show a strong association, but bedrooms and area are themselves correlated; the heatmap discloses this overlap and the user is warned against attributing a separate effect to each. In the third, a listing appears as an outlier; the user can toggle the flag and see whether summaries change materially, which distinguishes a data-entry error from a genuine premium property.

D. Handling Units and Locality Names

Two practical issues in listing data deserve explicit handling. First, area may be recorded in square feet, square yards, or square metres, and plots are commonly advertised in square yards in some markets; the tool requires a single unit per analysis and records the conversion if one is applied, because a mixed-unit price-per-area figure is meaningless. Second, locality names vary in spelling, abbreviation, and granularity (a neighbourhood, a zone, or a postal area). Names are normalized by a stored mapping, and records whose locality cannot be matched are reported rather than assigned. Both rules are part of the logged cleaning protocol so that a different analyst can reproduce the same grouping.

VI. ANALYSIS PLAN AND RESULT REPORTING

The numerical results of this study depend on the finalized listing dataset. To avoid asserting values that have not been computed, this section defines the tables the analysis will produce and the way each will be interpreted; the bracketed cells are to be populated directly from the analysis script output. No numerical finding is claimed in this version of the manuscript.

A. Descriptive Statistics

TABLE VI DESCRIPTIVE STATISTICS (N = [ ])

Variable	Mean	SD	Median	Min	Max
Price (INR)	[ ]	[ ]	[ ]	[ ]	[ ]
Area (sq ft)	[ ]	[ ]	[ ]	[ ]	[ ]
Price per sq ft	[ ]	[ ]	[ ]	[ ]	[ ]
Bedrooms	[ ]	[ ]	[ ]	[ ]	[ ]
Age (years)	[ ]	[ ]	[ ]	[ ]	[ ]

The tool reports the gap between mean and median price as an indication of skew, together with quartiles and the proportion of records flagged by the IQR rule. Because the central half of listings can span a wide price range, an overall average should not be used to infer the price of any individual property.

B. Correlation and Regression Summaries

TABLE VII PAIRWISE ASSOCIATIONS WITH PRICE

Pair	Pearson r	Spearman ρ	95% CI for r	p-value	R^2
Area vs. price	[ ]	[ ]	[ ]	[ ]	[ ]
Bedrooms vs. price	[ ]	[ ]	[ ]	[ ]	[ ]
Age vs. price	[ ]	[ ]	[ ]	[ ]	[ ]
Area vs. price per sq ft	[ ]	[ ]	[ ]	[ ]	[ ]
ln(area) vs. ln(price)	[ ]	[ ]	[ ]	[ ]	[ ]

Hedonic theory [1], [3] leads to the expectation that floor area is positively associated with price; this is a hypothesis to be examined, not a result. Bedrooms and area are typically correlated with each other, so the tool also displays the full correlation matrix to disclose such overlap rather than attributing a separate effect to each attribute. Every coefficient is displayed beside its scatter plot [8].

C. Locality and Property-Type Comparison

TABLE VIII LOCALITY SUMMARY (TEMPLATE)

Locality	n	Median price	Median INR/sq ft	IQR of INR/sq ft
[Locality 1]	[ ]	[ ]	[ ]	[ ]
[Locality 2]	[ ]	[ ]	[ ]	[ ]
[Locality 3]	[ ]	[ ]	[ ]	[ ]

Localities with few listings are labelled with their sample size and are not ranked, since a median from a handful of advertisements is unstable. Differences between locality medians are exploratory: they may reflect differences in property mix, age, or listing practices rather than differences in land value. Where a formal comparison across more than two groups is shown, it is reported as a descriptive check, with its assumptions stated.

D. Consistency Check Against Offline Computation

Once the dashboard is built, every summary card and chart is compared with the offline script. Any difference, such as a rounding variation or a different row-inclusion rule, is reported rather than silently reconciled. The tool's numbers are acceptable only if the data version and the formula are the same in both places.

VII. DISCUSSION

A. Answers to the Research Questions

RQ1 is answered by the audit and distribution outputs, which characterize the file but not any wider market. RQ2 is answered by the association tables, which describe linear and monotonic relationships and their uncertainty. RQ3 is answered through locality and property-type summaries, which are exploratory because listing mix is uncontrolled. RQ4 is answered by the traceability design and parity tests. RQ5 is answered by the prerequisites below, not by a numerical result.

B. Implications for Dashboard Design

First, metric definitions should be visible: tooltips or a methodology panel should state that price is an asking price, how price per square foot is computed, and which rows are included. Second, views should complement one another: a card communicates centre, a histogram spread, a scatter plot a relationship between two continuous variables, and a box plot a categorical split; no single chart answers all four. Third, limitations should be discoverable at the point of possible overinterpretation, since a note in a paper does not reach a user who only sees the interface. Fourth, outputs should be phrased as prompts for investigation, not as verdicts such as "overpriced." Fifth, every export should preserve the source date, definitions, and filters.

C. Implications for Research Practice

A small, interpretable analysis is a useful first step before machine learning. Descriptive methods reveal data-quality defects, unexpected ranges, and variable relationships that must be understood before prediction is attempted. Building a price predictor before checking duplicated advertisements, rounded prices, or target leakage could yield persuasive but untrustworthy accuracy figures. A future predictive extension should use a leakage-aware train/test design, appropriate error metrics such as median absolute percentage error, and subgroup analysis by locality, and it should be evaluated separately from the exploratory tool.

VIII. LIMITATIONS, ETHICS, AND FUTURE WORK

The main limitations are as follows. (1) Listing data contain asking prices, which can differ from transaction prices, and they are not a random sample of the market. (2) The dataset's provenance, collection date, and licence must be documented before any public release. (3) A single snapshot cannot establish temporal trends or causal relations. (4) Omitted variables such as exact micro-location, building quality, floor, legal status, and negotiation affect price and are absent or unverified. (5) Pearson and OLS summaries capture only linear patterns and rest on untested assumptions. (6) No user study, baseline comparison, or latency benchmark has yet been conducted.

On ethics, listings may contain seller names, contact details, or precise addresses; these are excluded from analysis and from reports, and only aggregates are published. Price statistics for small localities can disadvantage or stigmatize

neighbourhoods, so group sizes are shown and low-count groups are not ranked. The tool is decision support, not a valuation instrument, and should not be used alone for lending, taxation, or pricing decisions.

Future work includes documenting the data source, completing the analysis on the finalized dataset, adding a time dimension if dated listings are available, testing the dashboard with representative users on task completion and interpretation accuracy, and, separately, evaluating a predictive model with an independent held-out set.

IX. VERIFICATION AND EVALUATION PLAN

The current work is a design and analysis specification, not a completed software validation. Table IX translates the identified risks into verifiable activities for the next release, prioritized by consequence: numeric consistency and privacy are foundational, and usability and impact evaluation follow. Passing a functional test does not demonstrate that the tool improves anyone's decisions.

TABLE IX PROPOSED VERIFICATION PLAN

ID	Risk	Verification procedure
T-01	Different data versions	Freeze one file, record its checksum, and confirm the dashboard reads the same version
T-02	Arithmetic or rounding defects	Use hand-checkable fixture rows with known medians, ratios, and correlations
T-03	Wrong chart grouping	Test boundary values, empty localities, and single-record groups
T-04	Asking price mistaken for value	Ask users to interpret displays using standard scenarios; score errors
T-05	Correlation read as causation	Scenario-based interpretation test with a scoring rubric
T-06	Exposure of personal data	Confirm identifiers are excluded from views and exports
T-07	Accessibility or layout failures	Test keyboard use, contrast, and mobile widths

X. CONCLUSION

This paper presented the Real Estate Price Exploration Tool, an exploratory dashboard that brings data-quality auditing, derived price measures, correlation and regression summaries, and locality comparison into one interface. Grounded in hedonic pricing theory, the tool is deliberately descriptive: it helps users see how price varies with area, configuration, age, and location in a given listing file, while stating that asking prices are not values and associations are not causes. The methodology, traceability principles, result templates, and verification plan are specified so that the analysis can be completed and reproduced on the finalized dataset. Documenting the dataset, populating and checking the results, and evaluating the interface with users are the necessary next steps before any broader claim is made.

APPENDIX A: VARIABLE DICTIONARY

Table X records the expected fields and their treatment. The descriptions are operational interpretations of column names; documentation from the dataset source should supersede them if available.

TABLE X VARIABLE DICTIONARY (PROVISIONAL)

Field	Provisional role	Open definition questions
price	Outcome	Asking or sale price? Currency, taxes, and fees included?
area_sqft	Size attribute	Built-up, carpet, or super built-up area?
bedrooms, bathrooms	Configuration	Counting rules for studios and extra rooms
age_years	Condition proxy	Construction year or listing-based estimate?
locality	Location label	Granularity and naming standard
property_type	Category	Category definitions and overlaps
listing_date	Date	Posting or update date? Time zone

APPENDIX B: INTERPRETATION CHECKLIST

- A mean or median describes the analyzed listings; it does not give the price of any particular property.
- A positive correlation indicates positive association in this file; it does not show that adding area or a bedroom will raise price by a given amount.
- A small p-value does not imply practical importance and does not remove bias from an unrepresentative sample.
- An R^2 computed on the data used to fit a line is not a test-set accuracy statistic.
- A locality median based on few listings is unstable and should not be used to rank localities.
- Price per square foot depends on how area is defined; compare only like-for-like definitions.
- An outlier flag marks an unusual value for review; it is not a finding that the value is wrong.
- A working dashboard is evidence of a software artifact, not evidence of better decisions.

APPENDIX C: THREATS TO VALIDITY

Construct validity. Price, area, and age are interpreted from column names; asking price may not reflect market value, and area definitions may differ across listings. These gaps affect what the associations mean.

Internal validity. The analysis is observational and cross-sectional. Micro-location, building quality, floor level, legal status, and negotiation are absent or unverified, so fitted lines are descriptive and not causal estimates.

External validity. The source, sampling frame, city coverage, and collection date of the dataset determine to what extent results apply elsewhere. Listings over-represent properties that sellers choose to advertise online, so conclusions should not be extended to the wider market.

Statistical conclusion validity. Pearson correlation captures linear association only, p-values are exploratory with no multiplicity adjustment, and regression assumptions have not been fully tested. Log transformation changes the interpretation of slopes and must be reported with every fitted model.

Implementation validity. The dashboard has not been subjected to a formal software verification campaign. Differences between on-screen values and offline results,

such as rounding or row-inclusion rules, are possible and are why automated parity tests are proposed.

Conclusion validity. The study supports statements about descriptive patterns in the analyzed file and about the design scope of the prototype. It does not support claims of predictive validity, fairness, reduced search effort, or improved decisions; those require separate data and evaluation designs.

APPENDIX D: REPRODUCIBLE ANALYSIS SEQUENCE

- Load the listing file, preserving header names, and record its name and checksum.
- Verify row and column counts, data types, and numeric parsing of price and area.
- Count missing values, exact duplicates, duplicate identifiers, and near-duplicate advertisements.
- Apply logged cleaning rules (unit conversion, locality mapping, non-positive values) and record exclusions.
- Derive price per square foot and log price; flag IQR outliers without deleting them.
- Compute descriptive statistics, Pearson and Spearman coefficients, confidence intervals, and simple regressions.
- Aggregate by locality, property type, and bands; suppress or label small groups.
- Generate all tables and figures from the same data frame and store the library versions used.
- Compare offline outputs with the dashboard and document differences instead of editing expected values.

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